

BUSAN, Korea, Republic of

WMO: 471590

Lat: 35.1047N

Lon: 129.0319E

Elev: 71

StdP: 100.48

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.4	-21.6	0.5	-1.8	-19.5	0.7	-0.9	10.1	2.5	9.2	2.0	4.2	340	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	31.5	25.6	30.4	25.2	29.3	24.7	26.6	29.7	26.0	29.1	25.4	28.4	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.1	28.4	25.1	20.4	28.0	24.5	19.6	27.6	83.6	30.0	81.0	29.1	78.6	28.4	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.1	7.1	-8.2	33.1	2.5	1.6	-10.0	34.3	-11.4	35.2	-12.8	36.1	-14.6	37.3		
			-10.4	27.3	2.0	1.2	-11.9	28.1	-13.0	28.8	-14.1	29.5	-15.6	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.8	5.7	9.3	14.0	18.2	21.3	24.8	26.6	23.0	18.4	12.3	6.1	(6)
		DBStd	8.13	3.57	3.63	3.26	2.72	2.27	1.78	2.56	2.26	2.21	2.67	3.69	3.99	
		HDD10.0	527	193	126	52	3	0	0	0	0	1	20	131		
		HDD18.3	1843	451	355	280	131	2	0	0	0	31	182	381		
		CDD10.0	2473	1	5	31	123	253	338	459	515	389	262	89	8	
		CDD18.3	748	0	0	0	1	26	89	201	257	139	34	1	0	
		CDH23.3	4771	0	0	0	2	48	191	1443	2402	635	51	0	0	
		CDH26.7	1159	0	0	0	0	2	7	340	723	87	1	0	0	
(14)	Wind	WSAvg	3.4	3.5	3.5	3.8	3.8	3.5	3.2	3.6	3.5	3.3	3.1	3.1	3.3	(14)
(15)	Precipitation	PrecAvg	1499	31	45	85	144	151	200	279	230	174	70	59	28	(15)
(16)		PrecMax	2398	124	126	220	385	419	938	886	700	417	248	266	163	(16)
(17)		PrecMin	902	0	0	6	12	0	18	53	17	17	0	0	0	(17)
(18)		PrecStd	398	26	30	49	75	88	150	166	172	119	55	53	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.1	16.0	18.7	22.7	26.3	27.5	32.2	33.4	30.3	26.0	22.1	16.5		(19)
(20)		0.4% MCWB	10.2	11.0	12.0	14.7	17.2	20.9	25.9	26.0	24.8	19.4	16.1	11.8		(20)
(21)		2% DB	11.7	14.1	17.0	21.0	24.6	26.3	30.9	32.1	28.7	24.7	20.2	14.6		(21)
(22)		2% MCWB	7.1	9.7	11.3	14.0	16.7	20.4	25.5	25.7	23.4	18.6	14.9	10.2		(22)
(23)		5% DB	10.2	12.4	15.5	19.6	23.4	25.4	29.7	31.1	27.6	23.6	18.8	13.1		(23)
(24)		5% MCWB	5.9	7.9	10.6	13.5	16.6	20.0	25.2	25.4	22.4	18.1	14.2	8.5		(24)
(25)		10% DB	8.8	10.9	14.3	18.3	22.1	24.4	28.6	30.1	26.6	22.5	17.5	11.6		(25)
(26)		10% MCWB	4.6	6.6	9.9	13.0	16.1	19.7	24.7	25.1	21.9	17.4	13.2	7.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.8	13.7	14.7	18.0	20.3	23.7	27.4	27.7	26.0	22.4	18.1	13.3		(27)
(28)		0.4% MCDB	13.0	15.0	16.5	19.5	22.3	25.1	30.2	31.1	29.1	23.8	20.0	14.7		(28)
(29)		2% WB	8.4	11.0	13.3	16.3	19.2	22.7	26.5	26.7	24.9	20.7	16.5	11.1		(29)
(30)		2% MCDB	10.6	13.0	15.3	18.5	21.7	24.0	29.3	30.1	27.4	23.3	18.9	14.0		(30)
(31)		5% WB	6.8	8.5	12.0	15.3	18.4	21.9	25.8	26.2	24.0	19.4	15.1	9.4		(31)
(32)		5% MCDB	9.4	11.6	14.5	17.8	21.2	23.6	28.6	29.4	26.6	22.3	18.2	12.5		(32)
(33)		10% WB	5.4	7.0	10.4	14.3	17.6	21.1	25.2	25.7	22.8	18.2	13.8	7.9		(33)
(34)		10% MCDB	8.1	10.3	13.6	17.1	20.6	23.3	27.8	28.9	25.6	21.5	16.8	11.3		(34)
(35)	Mean Daily Temperature Range	MDBR	7.2	7.4	7.3	6.9	6.4	5.3	4.6	5.0	5.6	6.9	7.1	7.2		(35)
(36)		5% DB MCDBR	7.8	8.1	8.0	8.2	8.0	6.5	5.8	6.2	6.4	7.4	7.5	7.7		(36)
(37)		5% WB MCWBR	5.8	5.9	5.6	5.4	4.9	3.8	2.7	2.9	3.5	4.3	5.2	5.9		(37)
(38)		5% WB MCDBR	6.6	7.0	6.5	6.3	6.3	4.5	5.1	5.4	5.3	5.9	6.5	7.1		(38)
(39)		5% WB MCWBR	6.0	6.2	6.2	5.0	4.6	3.2	2.7	2.9	3.4	4.3	5.4	6.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.406	0.456	0.528	0.558	0.570	0.607	0.582	0.539	0.490	0.455	0.424	0.399		(40)
(41)		taud	2.099	1.998	1.839	1.821	1.846	1.833	1.958	2.075	2.156	2.132	2.131	2.132		(41)
(42)		Ebn at Noon	762	765	742	746	743	714	730	754	772	759	737	741		(42)
(43)		Edh at Noon	124	154	196	209	207	209	184	161	142	133	120	114		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.66	3.29	4.29	4.98	5.46	4.82	4.63	4.70	3.90	3.59	2.74	2.43		(44)
(45)		RadStd	0.21	0.37	0.39	0.39	0.59	0.44	0.71	0.57	0.41	0.34	0.33	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page